The Application of Blockchain in Face Recognition

目 录

CONTENTS.

01 Background

02 Literature Review

03 Theoretical Framework

04 Conclusion

PART 01

Background

01 BACKGROUND

PART 01.

- facial recognition technology has been extremely widely applied in numerous fields.
- with the increasingly widespread application of facial recognition technology, the issues of data security and privacy protection it faces have become more and more prominent.
- Characteristics of the Blockchain: Inability to Be Tampered With, Decentralization, Transparency, Anonymity.

PART 02

Literature Review

02

PART 02.

webs

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面向种质资源领域的人脸识别与区块链技术融合的身份认证研究

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摘要： 种质资源为战略性资源，种质资源平台能够有效解决种质数据的信息孤岛问题，实现资源共享。在种质资源领域内，传统的种质资源身份认证平台仅通过单一服务器进行认证，容易导致数据泄露或者丢失，在认证过程中用户和平台也面临着更多潜在的安全问题，如：中间人攻击问题、重放攻击问题和拒绝服务攻击问题。针对传统的种质资源平台所面临的上述安全性问题，本文提出了一种基于区块链和人脸识别技术的身份认证机制，该机制融合了区块链技术的去中心化、数据不可篡改和信息可追溯等特点，以及人脸识别技术在认证方面安全快捷、不易仿冒的突出优势，通过区块链的共识机制和智能合约的交互设计，有效解决了传统身份认证中面临的安全性问题。本文主要的研究内容和总结如下：（1）阐述了传统种质资源平台身份认证机制存在的安全性问题，分析并研究...

关键词： 种质资源; 身份认证; 区块链; 人脸识别; 智能合约

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PART 03

Theoretical Framework

algorithm

Proof of Work, PoW Practical

- calculate a valid nonce (random number) in the shortest possible time.
- However the computing power has been continuously concentrated, and the problem of computing power centralization has become increasingly severe.

Proof of Stake, PoS

- ensure the generation cycle of new blocks and enhances the stability of the entire system.
- There are multiple nodes competing for the right to develop the same block simultaneously, which creates a large number of hard forks and undermines the consistency.

Delegated Proof of Stake, DPOS

- reduces the number of nodes for verification and bookkeeping, shortening the time and significantly saving energy and hardware devices.
- the Witness mechanism leads to excessive centralization of computing power, leaving the **fairness** of the system unguaranteed.

Byzantine Fault Tolerance, PBFT

- adjusting the weights of nodes participating in the consensus
- conducting selective broadcasting, the invalid communication between nodes is reduced.

FaceNet Model

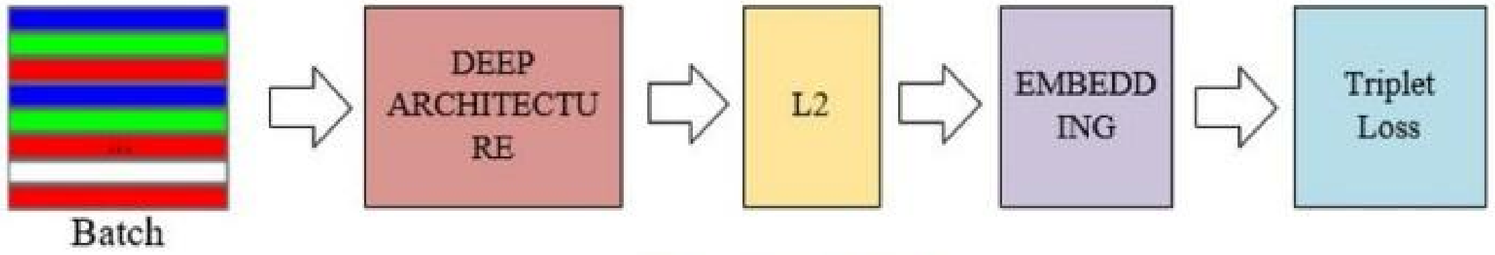

图 2.5 FaceNet 模型

Fig.2.5 FaceNet Model

Perform batch processing on images through Batch, and map faces to feature vectors in the Euclidean space via a deep Convolutional Neural Network (CNN).

The problem of decentralization

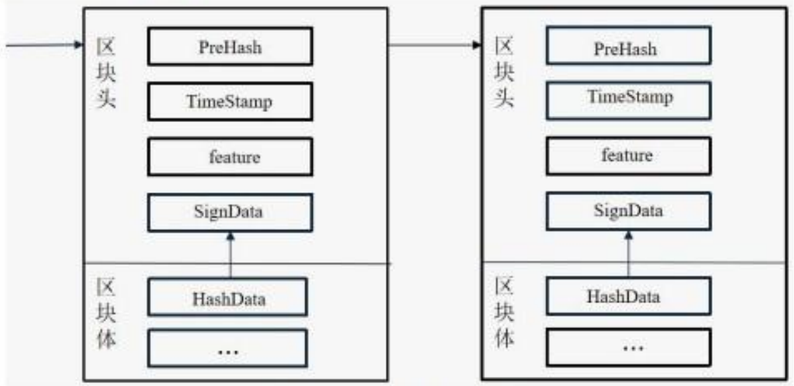

图 3.2 区块链数据结构

Fig.3.2 Data Structure of Blockchain

Hash calculation and signing are carried out through the Merkle algorithm, and the data is then packaged in the form of a smart contract and stored in the blockchain. Data is stored in a chain structure, and since each block is interrelated with the ones before and after it, can’t be easily change

PART 04
conclusion

04

PART 04.

plan

show more about its use by concerning about the securities in face recognizing

collect

collect datas about faces, detect them with this models

concern

whether it could recognize same faces in different situation

develop

apply it to Identity authentication design

make it wider to much accurate and much faster